Track : Spark AWS

Week-5 : Data ware Housing and Big Data

Day 1: Recap and Mocktest

Day 2: debriefing and recap

Day 3: practice

Day 4: Milestone-2

Day 5: Data Warehousing Fundamentals

Understanding traditional Relational Database Management Systems (RDBMS)

- Challenges encountered with RDBMS
- Issues related to scaling up in RDBMS
- Different RDBMS cluster approaches

Data Warehousing Architecture

- Introduction to Data Warehousing
- Data Warehousing Architecture Overview
- Components of Data Warehousing Architecture
- Importance of Data Warehousing in modern enterprises

Designing Data Warehouse and Introduction to Big Data

- Concepts of Fact Tables and Dimension Tables
- Designing Fact Tables and Dimensions
- ETL (Extract, Transform, Load) process in Data Warehousing
- Best practices in designing ETL pipelines

Big Data Introduction and Getting Started with Hadoop

- Overview of Big Data and its challenges
- Introduction to Hadoop as a Big Data solution
- Getting started with Hadoop
- Understanding the Hadoop framework

Hadoop Framework and HDFS

- Stepping into the Hadoop ecosystem

- Hadoop 2.x - YARN (Yet Another Resource Negotiator)
- Understanding the role of YARN in Hadoop
- Practical insights into Hadoop applications

HDFS (Hadoop Distributed File System)

- Introduction to HDFS
- Why HDFS is crucial for Big Data processing
- Working on HDFS - hands-on exercises
- Real-world applications of HDFS

Day 6: Big Data and Hadoop & HDFS As Data Lake

MapReduce Programming Paradigm

- Introduction to MapReduce
- Understanding the MapReduce programming paradigm
- Key concepts and principles behind MapReduce

Closer Look at MapReduce

- Deep dive into the components of MapReduce
- Exploring the Map phase and Reduce phase
- Hands-on examples of MapReduce programs

Practical Approach to MapReduce

- Real-world use cases of MapReduce
- Best practices and optimization techniques
- Challenges and solutions in implementing MapReduce

Understanding MapReduce with an Example

- Walkthrough of a practical MapReduce example
- Explanation of Map and Reduce functions
- Analyzing the output and results

Hadoop 1.x vs Hadoop 2.x

- Evolution of Hadoop from version 1.x to 2.x
- Key enhancements and features in Hadoop 2.x
- Impacts on MapReduce programming model

HDFS as a Data Lake

- Overview of Hadoop Distributed File System (HDFS)
- Utilizing HDFS as a Data Lake
- Storing and managing large-scale data in HDFS File Formats and Their Use Cases

- Introduction to various file formats (CSV, Parquet, ORC, Avro, etc.)
- Use cases for different file formats in Hadoop
- Choosing the right file format for specific scenarios

Hadoop Installation (Step by Step)

- Step-by-step guide to installing Hadoop
- Configuring Hadoop environment
- Verifying the installation and setting up a basic Hadoop cluster

Week-6

Day 1: Hive & Sqoop

Apache Hive Basics

- Introduction to Apache Hive
- Key features and advantages of using Hive
- Hive architecture and components

Hive Illustration - External Table in Hive (Partition & Bucketing)

- Understanding External Tables in Hive
- Implementing partitioning and bucketing for optimization
- Hands-on illustration with external tables

Hive Illustration - Loading Different File Formats (Serialization & Deserialization)

- Overview of file formats in Hive (CSV, Parquet, ORC, Avro)
- Serialization and deserialization in Hive
- Loading data in different file formats

Hive Illustration - Loading Data into Hive Tables

- Step-by-step process of loading data into Hive tables
- Handling various data sources and formats
- Best practices for efficient data loading

Hive Illustration - Simple Operations on Hive Table

- Performing basic operations on Hive tables (INSERT, UPDATE, DELETE)
- Data manipulation and transformation in Hive
- Practical examples of simple operations

Hive Illustration - Query Operations on Hive Table (RegEX)

- Introduction to Regular Expressions (RegEX) in Hive
- Query operations using RegEX on Hive tables
- Real-world scenarios and use cases

Hive Illustration - Querying Complex Structures

- Handling complex data structures in Hive (Arrays, Maps, Structs)
- Querying and processing nested data
- Advanced examples of querying complex structures

Hive Illustration - Views

- Creating and managing views in Hive
- Optimizing query performance using views
- Use cases and considerations for views in Hive

SQL using Hive and Impala

- Writing SQL queries in Hive
- Introduction to Impala for interactive SQL queries
- Comparative analysis of Hive and Impala

Joins in Hive

- Understanding different types of joins in Hive
- Implementing joins for combining data from multiple tables
- Optimizing join operations in Hive

Importing Data using Sqoop

- Introduction to Sqoop for data import/export
- Configuring Sqoop for Hive integration
- Importing data into Hive using Sqoop

Importing Data into Hive using Sqoop

- Step-by-step guide to importing data into Hive using Sqoop
- Handling data types and schema mapping
- Best practices for efficient data import

Day 2: Spark & PySpark + Milestone-2 RETEST

Getting Started - Spark Basics

- Introduction to Apache Spark
- Key features and advantages
- Spark ecosystem overview

Spark vs Hadoop

- Comparative analysis of Spark and Hadoop
- Use cases and scenarios for choosing Spark

Spark - Architecture

- High-level architecture of Apache Spark
- Components and their roles in Spark architecture

Driver Program

- Understanding the role of the Driver Program in Spark
- Interaction with the Spark cluster

Cluster Manager

- Overview of Cluster Managers in Spark (e.g., Apache Mesos, Apache YARN)
- Managing resources in a Spark cluster

Executor, Tasks

- Executor role in Spark
- Task execution in Spark clusters

Direct Acyclic Graph (DAG)

- Introduction to DAG in the context of Spark
- Understanding the execution flow in Spark

Spark Terminologies

- Explanation of key Spark terminologies
- Clarification of terms used in Spark ecosystem

Introduction to RDD

- Overview of Resilient Distributed Datasets (RDD) in Spark
- Characteristics and advantages of RDD

Pyspark - Hands-on

- Practical introduction to PySpark
- Hands-on exercises with PySpark basics

Pyspark - Clustering

- Introduction to clustering in PySpark
- Hands-on implementation of clustering algorithms in PySpark

Day 3: RDD & Spark Cluster Execution

Spark Program Lifecycle

- Overview of the lifecycle of a Spark program
- Understanding the stages of Spark program execution

RDDs - Building Blocks of Spark

- Introduction to Resilient Distributed Datasets (RDDs)
- Characteristics and benefits of using RDDs in Spark

RDD - Transformations

- Explanation of RDD transformations
- Commonly used transformations in Spark

RDD Operations - Actions

- Overview of RDD actions in Spark
- Examples of actions triggering the execution of Spark computations

RDD Lineage

- Understanding RDD lineage in Spark
- Fault tolerance mechanisms provided by RDD lineage

Concepts of Execution Methods - Lazy Execution, etc.

- Lazy execution in Spark and its benefits
- Strategies for optimizing Spark program execution

Data Pipeline

- Designing and implementing a data pipeline in Spark
- Connecting various stages in a Spark data pipeline

RDD from Python Collections

- Creating RDDs from Python collections
- Hands-on exercises for working with Python collections in Spark

RDD from File

- Loading data into Spark RDD from external files
- File formats and considerations for RDD creation

Scheduling Workflows using Oozie

- Introduction to Apache Oozie
- Designing and scheduling Spark workflows with Oozie

Day 4: Spark SQL

Introduction to Spark SQL

- Overview of Spark SQL
- Integration of SQL queries with Spark
- Key features and use cases of Spark SQL

Spark SQL as an ETL Tool

- Leveraging Spark SQL for Extract, Transform, Load (ETL) operations
- Designing and implementing ETL processes with Spark SQL

Spark SQL Performance Tuning

- Strategies for optimizing performance in Spark SQL
- Best practices for improving query performance

Dataframes from RDD

- Transformation of Resilient Distributed Datasets (RDDs) into DataFrames
- Benefits and characteristics of working with DataFrames in Spark

Dataframes from CSV

- Loading data from CSV files into Spark DataFrames
- Handling CSV-specific considerations and challenges

Spark Streaming and Real-Time Data Analytics

- Introduction to Spark Streaming for real-time data processing
- Use cases and scenarios for Spark Streaming

Spark Streaming Architecture

- Overview of the architecture of Spark Streaming
- Components and their roles in Spark Streaming applications

Real-Time Analytics - Get it with Twitter Demo

- Practical demonstration of real-time analytics using Twitter data

- Implementing and visualizing real-time analytics with Spark Streaming

Day 5: Kafka & Spark Streaming

- Introduction to Kafka - The Need for Kafka
- Understanding the challenges addressed by Apache Kafka
- Importance of real-time, distributed, and fault-tolerant data streaming
- Use cases and scenarios where Kafka excels
- Kafka Components
- Overview of key components in the Kafka ecosystem
- Broker, Producer, Consumer, Zookeeper, and other essential elements
- Roles and responsibilities of each component in a Kafka cluster
- Kafka Workflow
- Deep dive into the typical workflow of Kafka
- Producing and consuming messages
- Kafka topics and partitions
- Message serialization and deserialization
- Spark Streaming
- Introduction to Apache Spark Streaming
- Real-time data processing using micro-batch processing
- Integration of Spark Streaming with Kafka for scalable and fault-tolerant stream processing
- Use cases and benefits of Spark Streaming

Day 6 Introduction to Airflow

- Introduction to Airflow
- Overview of Apache Airflow
- Role in orchestrating complex workflows and data pipelines
- Key features and benefits of using Apache Airflow
- Airflow Architecture & Components
- Deep dive into the architecture of Apache Airflow
- Components such as Scheduler, Web Server, Metadata Database, and Executor
- Understanding the flow of task execution in Airflow
- Airflow Operators
- Introduction to Airflow Operators
- Pre-built tasks for common operations
- Extending and creating custom operators
- Creating Your First Workflow
- Step-by-step guide to creating a simple DAG (Directed Acyclic Graph)

- Defining tasks and dependencies
- Executing and monitoring the workflow
- How Time Works in Airflow
- Understanding the concept of time in Airflow
- Scheduling and triggering tasks based on time intervals
- Timezones and daylight saving considerations
- Overview of UI
- Navigating the Airflow web-based user interface
- Monitoring and visualizing DAGs and task runs
- Troubleshooting and logging features
- Interacting with the Cloud Platform
- Integration of Apache Airflow with cloud services
- Use cases and benefits of running Airflow on cloud platforms
- Scheduling Tasks Using CronTab
- Configuring task schedules with Cron expressions
- Fine-tuning scheduling parameters for tasks
- Creating an Airflow Instance
- Setting up and configuring an Apache Airflow instance
- Configuring connections and variables
- Best practices for Airflow instance management
- Pushing a DAG into Production
- Strategies for deploying Airflow DAGs to production
- Versioning and managing DAGs in a production environment
- Scaling and optimizing Airflow for production workloads

Week-7

Day 1: Advance Apache Airflow

- Airflow UI
  - In-depth exploration of the Airflow web-based user interface
  - Navigating through DAGs, tasks, and logs
  - Utilizing the UI for monitoring and managing workflows
- Reading Logs
  - Understanding how logs are generated and stored in Airflow
  - Reading and interpreting task logs for troubleshooting
  - Leveraging logging features for debugging workflows
- Task Actions
  - Exploring actions that can be performed on tasks
  - Retry, mark success, and other task actions
  - Managing tasks through the Airflow UI
- Deleting a DAG
  - Best practices and considerations for deleting a DAG
  - Steps to safely remove a DAG from the Airflow instance
  - Impact of DAG deletion on running workflows
- Xcoms in Airflow
  - Introduction to Xcoms (Cross-Communication) in Airflow
  - Sharing data between tasks within a DAG
  - Use cases and examples of using Xcoms
- Implementing Xcoms
  - Practical implementation of Xcoms in Airflow
  - Configuring tasks to exchange data using Xcoms
  - Handling data dependencies between tasks
- Variables in Airflow
  - Overview of Airflow Variables
  - Storing and managing dynamic values
  - Using variables for parameterizing workflows
- Dummy Operator
  - Understanding the Dummy Operator in Airflow
  - Placeholder tasks for creating workflow structure
  - Use cases and scenarios for employing the Dummy Operator
- Hooks
  - Introduction to Airflow Hooks
  - Interfacing with external systems and databases
  - Creating custom hooks for specific integrations

- Pools
    - Overview of Airflow Pools
    - Resource management and task concurrency control
    - Configuring and utilizing pools for workload optimization
- Airflow Sensors
    - Introduction to Airflow Sensors
    - Wait for external events before task execution
    - Use cases and practical implementation of sensors
- Backfill & Catchup
    - Exploring backfilling and catch-up options in Airflow
    - Re-running or scheduling past DAG runs
    - Managing historical workflow data
- Branching
    - Implementing conditional branching in Airflow
    - Dynamically choosing different paths based on task outcomes
    - Building flexible and dynamic workflows with branching

Day 2: Scala Introduction

- Variables, Methods, Classes, and Objects
    - Introduction to basic concepts in Scala programming
    - Declaration and usage of variables
    - Defining methods and functions
    - Creating classes and objects in Scala
- Package and Package Objects
    - Overview of packages in Scala
    - Organizing code into packages
    - Utilizing package objects for shared functionality
- Higher-Order Functions in Scala
    - Understanding higher-order functions
    - Passing functions as arguments
    - Returning functions as results
    - Practical examples of higher-order functions in Scala
- Scala Collections
    - Introduction to Scala collections
    - Immutable and mutable collections
    - Common operations on collections
    - Selection of appropriate collection types for specific use cases
- SBT (Scala Build Tool)
    - Overview of SBT as a build tool for Scala projects
    - Project structure and configuration in SBT

    - Managing dependencies with SBT
    - Building, testing, and running Scala applications using SBT

Day 3: Data Management Overview

- Overview of Data Management
    - Definition and significance of data management
    - Evolution of data management in enterprises
    - Challenges and opportunities in modern data management
- Importance of Data Management at Enterprises
    - Understanding the critical role of data in business operations
    - Impact of effective data management on decision-making
    - Compliance and regulatory considerations
- Multidimensional Data Representation and Manipulation
    - Exploring multidimensional data models
    - Manipulating data in multiple dimensions
    - Use cases and benefits of multidimensional data representation
- Design Practices and Methodologies
    - Overview of design practices for efficient data management
    - Methodologies for designing scalable and maintainable data solutions
    - Incorporating best practices in data management design
- Understanding Data Warehouse, Data Lake, Lake House
    - Differentiating between data warehouse, data lake, and lake house
    - Use cases and scenarios for each data storage architecture
    - Integration and interoperability between these data storage solutions
- Data Management Architecture
    - Building a comprehensive data management architecture
    - Components and layers of a robust data architecture
    - Scalability, flexibility, and performance considerations
- Quality Check - How to Build a Quality Check Pipeline
    - Implementing data quality checks in a data pipeline
    - Setting up validation and verification processes
    - Ensuring data quality throughout the data lifecycle
- Governance - How to Create Governance Layer in Pipeline
    - Establishing data governance principles
    - Designing a governance layer within a data pipeline
    - Ensuring compliance, security, and accountability
- Migration - What Are the Checklists for Migration
    - Checklist for planning and executing data migration
    - Ensuring data integrity during migration
    - Mitigating risks and challenges in the migration process

- Data Security - How to Track Security and Lineage
    - Strategies for ensuring data security in a data management system
    - Implementing security measures at various layers
    - Tracking data lineage for transparency and auditability

Day 4: Software Engineering Essentials

- Event-Based Spark Streams
    - Understanding event-based stream processing in Apache Spark
    - Real-time data processing with Spark Streaming
    - Use cases and scenarios for event-based streaming in Spark
- Stream Processing with Flink
    - Introduction to Apache Flink for stream processing
    - Data streams and their characteristics
    - Time-based window operators for processing streaming data
    - Stateful operators in Flink
    - Reading and writing from/to external systems in Flink
- Debugging in Stream Processing
    - Strategies for debugging streaming applications
    - Identifying and resolving common issues in real-time data processing
    - Debugging tools and techniques for Spark and Flink
- Docker and Kubernetes
    - Introduction to containerization with Docker
    - Orchestrating containers with Kubernetes
    - Benefits of containerization and orchestration in stream processing
- Microservices (Only Theory)
    - Overview of microservices architecture
    - Decoupling and scaling components in a microservices environment
    - Considerations for building microservices in a streaming data ecosystem
- FAST API (Document and Ready Demo)
    - Introduction to FAST API for building APIs in Python
    - Documenting APIs with Swagger

Day 5: Cloud Fundamentals

The Classical Enterprise

- Understanding the traditional enterprise model
- Characteristics and challenges of classical enterprises

Myths of Cloud Computing

- Debunking common misconceptions about cloud computing
- Clarifying misconceptions related to security, reliability, and cost

Service Delivery Models

- Overview of cloud service delivery models (IaaS, PaaS, SaaS)
- Choosing the right service model for specific business needs

Degree of Abstraction

- Understanding the level of abstraction in cloud services
- Benefits and considerations of abstraction in the cloud

Cloud Attributes

- Key attributes defining cloud computing (scalability, flexibility, etc.)
- Impacts of cloud attributes on business operations

Service Offerings

- Exploring various cloud service offerings
- Infrastructure, platform, and software services in the cloud

Managed Services

- Introduction to managed services in the cloud
- Outsourcing operational responsibilities to cloud providers

Cost Economics

- Cost considerations in cloud computing
- Analyzing the economics of cloud services

Elasticity

- Understanding elasticity in the context of cloud computing
- Scaling resources based on demand and workload

Virtualization

- Overview of virtualization technology in the cloud
- Benefits and use cases of virtualization

Evolving to PaaS

- Transitioning from traditional models to Platform as a Service (PaaS)

- Advantages and challenges of adopting PaaS

VM vs Containers

- Comparison between virtual machines (VMs) and containers
- Use cases and considerations for VMs and containers

Cloud Service Taxonomy

- Categorizing cloud services based on functionality
- Understanding the taxonomy of cloud offerings

Overview of Infrastructure Automation

- Introduction to infrastructure automation in the cloud
- Tools and practices for automating infrastructure deployment

Provisioning Process

- Step-by-step guide to provisioning resources in the cloud
- Automation of resource provisioning for efficiency

Sample Cloud Architecture - How It All Comes Together

- Building a sample cloud architecture
- Integration of services and components for a comprehensive solution

Day 6: AWS Compute & Storage

EC2

- Introduction to Amazon Elastic Compute Cloud (EC2)
- Overview of virtual servers in the cloud
- Use cases and benefits of EC2 instances

Load Balancing

- Understanding load balancing in the cloud
- Benefits of load balancing for distributing workloads
- Implementing load balancing with Amazon ELB (Elastic Load Balancer)

Autoscaling

- Introduction to autoscaling in cloud environments

- Dynamic scaling based on demand
- Implementation of autoscaling with Amazon Auto Scaling

Identity Management (IAM)

- Overview of AWS Identity and Access Management (IAM)
- Role of IAM in managing user access and permissions
- Implementing IAM policies for security and compliance

S3 - Deep Dive (Storage, Security, Archival, Access)

- In-depth exploration of Amazon Simple Storage Service (S3)
- Storage options and classes in S3
- Security features and best practices
- Archiving and retrieving data with S3
- Configuring access control and permissions in S3

EFS

- Introduction to Amazon Elastic File System (EFS)
- Scalable file storage in the cloud
- Use cases and benefits of EFS
- Implementation and management of EFS

EBS

- Overview of Amazon Elastic Block Store (EBS)
- Persistent block storage for EC2 instances
- Different types of EBS volumes
- Managing and optimizing EBS for performance

Week-8

Day 1: AWS Networking & Managed Services

VPC (Virtual Private Cloud)

- Introduction to Amazon Virtual Private Cloud (VPC)
- Network isolation and segmentation in AWS
- Components of VPC, including subnets and route tables
- Configuring and managing VPC for secure and scalable architecture

Security Groups

- Understanding AWS Security Groups

- Role in controlling inbound and outbound traffic to AWS resources
- Implementation and management of Security Groups

NACLs (Network Access Control Lists)

- Overview of Network Access Control Lists
- Controlling traffic at the subnet level
- Comparisons between Security Groups and NACLs

Bastion Hosts

- Role of Bastion Hosts in secure access to private instances
- Configuring and deploying Bastion Hosts
- Best practices for managing and securing Bastion Hosts

Route 53

- Overview of Amazon Route 53
- DNS (Domain Name System) management and routing
- Implementation of Route 53 for domain registration and routing policies

SNS (Simple Notification Service) and SQS (Simple Queue Service)

- Introduction to SNS for pub/sub messaging
- Overview of SQS for reliable message queuing
- Use cases and integration of SNS and SQS in distributed systems

CloudWatch

- Overview of Amazon CloudWatch
- Monitoring and management of AWS resources
- Creating alarms and dashboards for operational insights

Kinesis

- Introduction to Amazon Kinesis
- Real-time data streaming and processing
- Use cases and benefits of Kinesis streams and analytics

CloudTrail

- Understanding AWS CloudTrail
- Logging and monitoring API activity in AWS
- Compliance and security benefits of CloudTrail

X-ray + Insights

- Overview of AWS X-Ray for distributed tracing
- Usage of X-Ray for identifying and resolving performance issues
- CloudWatch Insights for log analysis and troubleshooting

RDS (Relational Database Service)

- Introduction to Amazon RDS
- Managed relational database service in the cloud
- Deployment, scaling, and management of RDS instances

Athena

- Overview of Amazon Athena
- Serverless query service for analyzing data in Amazon S3
- Writing and executing queries using SQL in Athena

Day 2: AWS - Serverless, NoSQL & Security

AWS Lambda

- Introduction to AWS Lambda
- Serverless computing and event-driven architecture
- Use cases and benefits of AWS Lambda functions
- Creating and deploying Lambda functions

API Gateway

- Overview of Amazon API Gateway
- Building and managing APIs in the cloud
- Integration with Lambda functions and other AWS services
- Securing and monitoring APIs with API Gateway features

DynamoDB

- Introduction to Amazon DynamoDB
- NoSQL database service in AWS
- Key-value and document data modeling
- Provisioned and on-demand capacity modes
- Best practices for designing and querying DynamoDB tables

Encryption - SDK and CLI

- Encryption options in AWS
- Using AWS SDKs for implementing encryption

- Command Line Interface (CLI) commands for encryption tasks
- Best practices for securing data with encryption

AWS SSM (Systems Manager)

- Overview of AWS Systems Manager
- Centralized management and automation of resources
- Features like Parameter Store, Run Command, and Session Manager
- Use cases and implementation of SSM for operational efficiency

Day 3: AWS Data Services

Big Data Processing with AWS

AWS EMR (Elastic MapReduce) - Deep Dive

- Overview of Amazon EMR
- Managed big data processing using Apache Hadoop and Apache Spark
- Architecture and components of EMR
- Configuring and managing EMR clusters
- Use cases and best practices for big data processing with EMR

AWS Redshift - Deep Dive

- Introduction to Amazon Redshift
- Managed data warehouse service in the cloud
- Columnar storage and parallel processing
- Designing and optimizing Redshift clusters
- Integration with other AWS services

Analytics and Data Visualization

AWS QuickSight

- Overview of Amazon QuickSight
- Business Intelligence (BI) service in the cloud
- Creating and visualizing dashboards
- Integration with various data sources

AWS Athena - Deep Dive

- Introduction to Amazon Athena
- Serverless query service for analyzing data in Amazon S3
- Querying data using SQL in Athena

- Performance optimization and best practices

Day 4: AWS Data Services - 2

Data Integration and ETL

AWS Glue - Deep Dive

- Overview of AWS Glue
- Fully managed extract, transform, and load (ETL) service
- Crawling, cataloging, and transforming data in the cloud
- Integration with other AWS services for data processing

Apache NiFi with AWS

- Introduction to Apache NiFi
- Open-source data integration and automation tool
- Use cases and benefits of NiFi in data workflows
- Integration of Apache NiFi with AWS services
- Designing and implementing data flows with NiFi in the AWS environment

Day 5: AWS Data Services – 3

Day 6: Recap and Revision

Week-9

Day-1 : Recap and Revision

Day-2 : Mock client Review

Day-3 : Milestone-3

Day-4 : capstone project

Day-5 : capstone project

Day-6 : capstone project

Week-10

Day-1 : capstone project + Milestone-3 Re test

Day-2 : capstone project

Day-3 : capstone project

Day-4 : Milestone – 4 capstone project Evaluation